

An Analysis of the Second-Order Taylor Approximation of the Pythagorean Expectation
Win-Loss Model
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Abstract

We extend the analysis performed by Dayaratna and Miller (2012) to the second-order Taylor approximation of the Pythagorean expectation model. We show that the second-order approximation performs worse than the first-order approximation on Major League Baseball (MLB) data from 2006 to 2025. We reproduce Dayaratna and Miller's first-order analysis on the data. We derive and perform the same analysis on the second-order approximation. Finally, we analyze the approximation accuracy of the second-order approximation compared to that of the first-order approximation.

Introduction

The Pythagorean expectation model was created by Bill James in 1980 to calculate the percentage of games that a baseball team is expected to win ($E[W]$) given the number of runs that the team has scored (RS) and the number of runs that the team has allowed (RA) (Dayaratna and Miller 2012):

$$E[W] = \frac{RS^\gamma}{RS^\gamma + RA^\gamma} \quad (1)$$

An alternative model was introduced by Michael Jones and Linda Tappin in 2005 (Dayaratna and Miller 2012):

$$E[W] = \frac{1}{2} + \beta(RS - RA) \quad (2)$$

This model simplifies James' nonlinear two-variable equation into a linear single-variable equation, the variable being the team's run differential ($RS - RA$).

A subsequent study by Kevin Dayaratna and Steven Miller (2012) showed that the Jones and Tappin model is a first-order Taylor approximation of the original model by expanding at the league average for runs scored and allowed (R_{avg} , R_{avg}) to arrive at:

$$E[W] \approx \frac{1}{2} + \frac{\gamma}{4R_{avg}} (RS - RA) \quad (3)$$

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Data

The data consists of runs scored, runs allowed, wins, and losses for MLB teams from the 20 most recent complete MLB seasons (2006–2025). This time period represents the current era of MLB. Note that the data consists of run totals, while the data that Dayaratna and Miller (2012) used consisted of run averages for the 1991 to 2011 MLB seasons.

The Dayaratna and Miller Analysis

To support their derivation of the first-order Taylor approximation of the Pythagorean expectation model, Dayaratna and Miller (2012) used the method of least squares to fit the following model on their data:

$$E[W] = \alpha + \beta(RS - RA) \quad (4)$$

We do the same for each season from 2006 to 2025. Below are the model estimates and statistics:

season	$\hat{\alpha}$	p-value for $\hat{\alpha}$	$\hat{\beta}$	p-value for $\hat{\beta}$	R_{avg}	$\hat{y}$	95% Lower Bound on $\hat{y}$	95% Upper Bound on $\hat{y}$	R^2
2006	0.500	7.49E-38	6.03E-04	2.98E-12	786.63	1.896	1.563	2.229	0.829
2007	0.500	4.40E-38	5.26E-04	1.90E-11	777.40	1.637	1.325	1.949	0.805
2008	0.500	1.44E-37	6.42E-04	4.24E-13	752.83	1.934	1.621	2.247	0.851
2009	0.500	4.25E-37	6.57E-04	5.35E-13	747.30	1.962	1.642	2.283	0.849
2010	0.500	3.28E-44	5.75E-04	1.03E-19	710.27	1.634	1.489	1.780	0.950
2011	0.500	7.76E-38	6.39E-04	9.27E-14	693.60	1.773	1.503	2.042	0.866
2012	0.500	6.09E-39	6.94E-04	2.06E-15	700.57	1.944	1.690	2.197	0.898
2013	0.500	6.22E-39	6.33E-04	1.07E-15	675.17	1.709	1.492	1.926	0.903
2014	0.500	3.84E-39	6.79E-04	5.91E-13	658.70	1.788	1.494	2.081	0.848
2015	0.500	6.21E-36	6.01E-04	9.52E-11	688.23	1.653	1.315	1.992	0.781
2016	0.500	5.30E-37	5.79E-04	3.75E-12	724.80	1.678	1.380	1.976	0.826
2017	0.500	1.72E-39	5.42E-04	1.51E-15	752.73	1.631	1.421	1.841	0.900
2018	0.500	4.74E-37	5.99E-04	5.28E-16	721.00	1.728	1.514	1.941	0.907
2019	0.500	5.54E-40	6.11E-04	6.18E-20	782.23	1.910	1.743	2.077	0.951
2020	0.500	6.63E-34	1.62E-03	3.49E-13	278.13	1.807	1.517	2.097	0.853
2021	0.500	5.83E-36	5.86E-04	9.14E-15	733.67	1.721	1.482	1.959	0.887
2022	0.500	1.58E-40	6.07E-04	1.55E-19	693.90	1.685	1.532	1.838	0.948
2023	0.500	2.29E-37	5.46E-04	5.47E-15	747.73	1.634	1.413	1.856	0.891
2024	0.500	7.75E-41	6.19E-04	7.00E-18	711.43	1.762	1.578	1.946	0.932

2025	0.500	3.09E-37	5.28E-04	1.88E-13	720.47	1.522	1.284	1.760	0.859
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Table 1: First-order Coefficients and Model Statistics

After performing Bonferroni corrections, the significance level is reduced from 0.05 to 0.0025. All of the coefficient estimates are significant. After using R_{avg} to calculate $\hat{\gamma}$ according to the relationship between $\hat{\beta}$ and $\hat{\gamma}$ shown in (3), the expected values for $\hat{\gamma}$ were recovered. The expected values for $\hat{\alpha}$ were also recovered.

The accepted value of γ that Dayaratna and Miller (2012) used was 1.82. This value falls inside the confidence intervals for every season except for 2010, as Dayaratna and Miller found, and 2025. Thus, Dayaratna and Miller's analysis and findings hold for seasons outside of those represented in their data.

Second-order Taylor Approximation Derivation

We now derive the second-order term of the second-order Taylor approximation. Let

$$f(x, y) = \frac{x^\gamma}{x^\gamma + y^\gamma} \quad (5)$$

for some value of γ . The second-order term of the Taylor approximation has the form

$$\frac{1}{2}f_{xx}(a, b)(x - a)^2 + 2f_{xy}(a, b)(x - a)(y - b) + f_{yy}(a, b)(y - b)^2 \quad (6)$$

where (a, b) is the expansion point. Dayaratna and Miller (2012) showed that the first-order partial derivatives of (5) are

$$f_x = \frac{\gamma x^{\gamma-1} y^\gamma}{(x^\gamma + y^\gamma)^2} \quad (7a)$$

$$f_y = -\frac{\gamma x^\gamma y^{\gamma-1}}{(x^\gamma + y^\gamma)^2} \quad (7b)$$

Taking the appropriate partial derivatives of (7a) and (7b), we get

$$f_{xx} = \frac{\gamma x^{\gamma-2} y^\gamma [(\gamma-1)y^\gamma - (\gamma+1)x^\gamma]}{(x^\gamma + y^\gamma)^3} \quad (8a)$$

$$f_{yy} = \frac{\gamma x^\gamma y^{\gamma-2} [(\gamma+1)y^\gamma - (\gamma-1)x^\gamma]}{(x^\gamma + y^\gamma)^3} \quad (8b)$$

$$f_{xy} = \frac{\gamma^2 x^{\gamma-1} y^{\gamma-1} (x^\gamma - y^\gamma)}{(x^\gamma + y^\gamma)^3} \quad (8c)$$

Evaluating (8a), (8b), and (8c) at expansion point (R_{avg} , R_{avg}) gives

$$f_{xx}(R_{avg}, R_{avg}) = -\frac{\gamma}{4R_{avg}^2} \quad (9a)$$

$$f_{yy}(R_{avg}, R_{avg}) = -\frac{\gamma}{4R_{avg}^2} \quad (9b)$$

$$f_{xy}(R_{avg}, R_{avg}) = 0 \quad (9c)$$

Plugging (9a), (9b), (9c) into (6) with $x = RS$ and $y = RA$ gives the second-order term as

$$-\frac{\gamma}{8R_{avg}^2} [(RS - R_{avg})^2 + (RA - R_{avg})^2] \quad (10)$$

Adding this term to Dayaratna and Miller's (2012) first-order result shown in (3), we arrive at the full second-order Taylor approximation of the Pythagorean expectation model:

$$E[W] \approx \frac{1}{2} + \frac{\gamma}{4R_{avg}} (RS - RA) - \frac{\gamma}{8R_{avg}^2} [(RS - R_{avg})^2 + (RA - R_{avg})^2] \quad (11)$$

Empirical Analysis

We perform Dayaratna and Miller's (2012) analysis on the second-order model by using least squares to fit the following model to the data:

$$E[W] = \alpha + \beta_1(RS - RA) - \beta_2[(RS - R_{avg})^2 + (RA - R_{avg})^2] \quad (12)$$

The goal is to see whether the estimated values of α match the corresponding values from Table 1, and whether the expected values of γ can be extracted from β_1 and β_2 .

Below are the model estimates and statistics:

season	$\hat{\alpha}$	p-value for $\hat{\alpha}$	$\hat{\beta}_1$	p-value for $\hat{\beta}_1$	$\hat{\beta}_2$	p-value for $\hat{\beta}_1$	R_{avg}	$\hat{\gamma}_1$	95% Lower Bound on $\hat{\gamma}_1$	95% Upper Bound on $\hat{\gamma}_1$	$\hat{\gamma}_2$	95% Lower Bound on $\hat{\gamma}_2$	95% Upper Bound on $\hat{\gamma}_2$	R^2
2006	0.498	2.22E-32	6.07E-04	1.18E-11	-2.18E-07	0.756	786.63	1.909	1.559	2.259	-1.078	-8.118	5.963	0.830
2007	0.507	5.36E-32	5.40E-04	2.20E-11	7.87E-07	0.210	777.40	1.680	1.364	1.996	3.805	-2.269	9.879	0.816
2008	0.500	5.01E-33	6.42E-04	1.85E-12	1.66E-08	0.968	752.83	1.933	1.607	2.259	0.075	-3.782	3.933	0.851
2009	0.495	7.41E-30	6.52E-04	1.52E-12	-5.07E-07	0.468	747.30	1.949	1.623	2.275	-2.264	-8.572	4.044	0.852
2010	0.500	2.28E-37	5.75E-04	6.16E-18	1.03E-08	0.975	710.27	1.633	1.469	1.798	0.042	-2.711	2.794	0.950
2011	0.507	1.42E-31	6.55E-04	1.68E-13	6.20E-07	0.220	693.60	1.817	1.540	2.093	2.387	-1.513	6.286	0.874
2012	0.502	4.07E-32	6.85E-04	5.46E-14	2.63E-07	0.660	700.57	1.920	1.641	2.199	1.031	-3.720	5.783	0.899
2013	0.512	2.00E-34	6.32E-04	1.66E-16	1.26E-06	0.010	675.17	1.708	1.512	1.904	4.607	1.173	8.042	0.924
2014	0.511	3.82E-35	6.70E-04	1.22E-13	1.44E-06	0.010	658.70	1.764	1.499	2.029	5.009	1.282	8.736	0.881
2015	0.502	3.05E-31	6.03E-04	2.21E-10	2.10E-07	0.737	688.23	1.661	1.313	2.009	0.797	-4.017	5.612	0.782

2016	0.506	1.67E-32	5.82E-04	4.64E-12	5.97E-07	0.241	724.80	1.688	1.391	1.984	2.510	-1.783	6.802	0.835
2017	0.495	1.61E-32	5.30E-04	3.40E-14	-4.20E-07	0.408	752.73	1.596	1.369	1.824	-1.904	-6.546	2.739	0.903
2018	0.511	1.08E-31	5.96E-04	4.59E-16	8.74E-07	0.071	721.00	1.718	1.513	1.923	3.635	-0.327	7.596	0.918
2019	0.505	2.09E-33	6.08E-04	2.66E-19	2.72E-07	0.365	782.23	1.901	1.732	2.070	1.331	-1.634	4.295	0.953
2020	0.491	7.31E-27	1.59E-03	3.00E-12	-3.37E-06	0.326	278.13	1.764	1.459	2.068	-2.088	-6.367	2.191	0.858
2021	0.508	5.51E-31	5.82E-04	1.45E-14	6.35E-07	0.167	733.67	1.709	1.474	1.944	2.734	-1.217	6.684	0.895
2022	0.500	3.37E-35	6.07E-04	7.87E-19	3.01E-08	0.920	693.90	1.686	1.529	1.842	0.116	-2.223	2.454	0.948
2023	0.503	6.23E-32	5.41E-04	4.19E-14	2.12E-07	0.598	747.73	1.619	1.386	1.852	0.949	-2.703	4.601	0.892
2024	0.505	4.20E-36	5.97E-04	9.16E-16	4.21E-07	0.210	711.43	1.699	1.490	1.907	1.706	-1.019	4.431	0.936
2025	0.501	7.50E-33	5.21E-04	2.76E-10	7.53E-08	0.834	720.47	1.501	1.183	1.819	0.313	-2.725	3.351	0.860

Table 2: Second-order Coefficients and Model Statistics

After performing Bonferroni corrections, the significance level is again reduced from 0.05 to 0.0025. None of the second-order coefficients were significant, while the estimates for $\hat{\alpha}$ and $\hat{\beta}_1$ are nearly identical to their counterparts for the first-order model. The calculated values of $\hat{\gamma}_1$ also agree with the values of γ for the first-order model, while the expected values of $\hat{\gamma}_2$, which should agree with the values of $\hat{\gamma}_1$, could not be recovered from the estimates of $\hat{\beta}_2$.

Approximation Evaluation

We now compare the second-order approximation shown in (11) with the first-order approximation shown in (3) to evaluate how accurate each approximation is to the original Pythagorean expectation model. The following plot of the three models together provides a visual sense of their relationship:

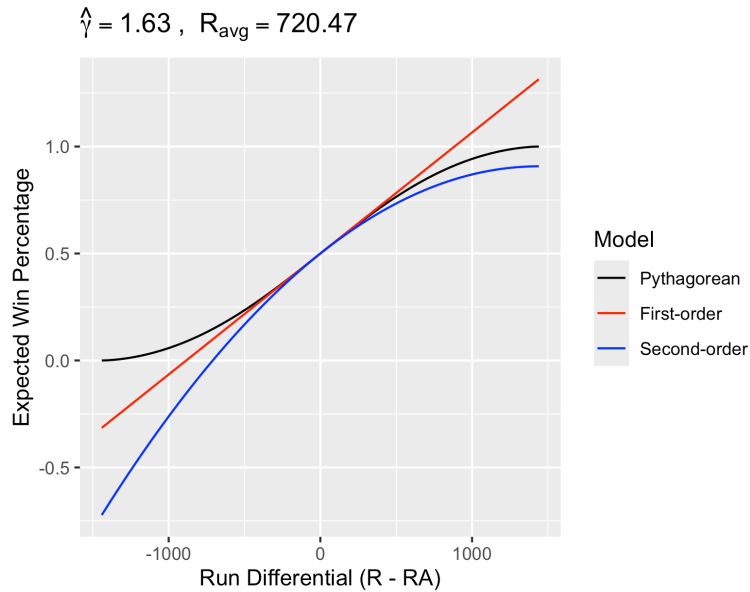


Figure 1: Model Curves

The first-order approximation underestimates the Pythagorean expectation values for $R < RA$ and overestimates for $R > RA$, while the second-order approximation consistently underestimates. Although Figure 1 shows the curves for only the 2025 season, the relationship between the curves is the same for every season.

We use nonlinear least squares to estimate the values of γ for the original model (see Appendix A). For each season, the values of $\hat{\gamma}$ and the data are plugged into (1), (3), and (11) to calculate the expected win percentages according to the Pythagorean expectation model and the win percentages approximated by the first- and second-order approximations. We then calculate the approximation error for the first- and second-order approximations, as well as the percent increase in the error of the second-order approximation with respect to the first-order approximation:

season	First-order Approximation Error	Second-order Approximation Error	Percent Change
2006	3.49E-03	6.33E-03	81.14
2007	3.38E-03	5.00E-03	48.14
2008	3.48E-03	8.95E-03	157.39
2009	4.54E-03	6.89E-03	51.95
2010	4.68E-03	7.03E-03	50.12
2011	5.87E-03	8.35E-03	42.25
2012	4.82E-03	1.02E-02	110.89
2013	5.30E-03	7.76E-03	46.30
2014	3.25E-03	7.36E-03	126.76
2015	4.82E-03	7.60E-03	57.81
2016	3.82E-03	8.21E-03	114.98
2017	3.75E-03	7.31E-03	95.03
2018	4.27E-03	9.00E-03	111.04
2019	5.64E-03	1.09E-02	93.47
2020	6.85E-03	1.31E-02	90.97
2021	4.98E-03	1.08E-02	117.53
2022	4.69E-03	1.08E-02	130.72
2023	5.07E-03	9.07E-03	78.80
2024	5.10E-03	9.52E-03	86.66
2025	7.05E-03	1.51E-02	114.26

Table 3: Approximation Errors

Unexpectedly, the first-order approximation is more accurate than the second-order approximation for every season, with the errors for the second-order approximation reaching more than double that of the first-order approximation for some seasons. To understand this, we plot the errors of both approximations against the Euclidean distances of each team from the expansion point:

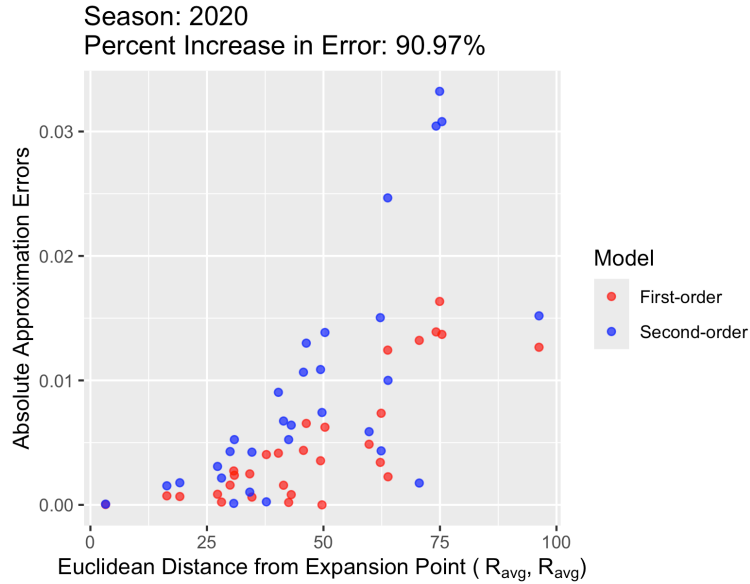


Figure 2: Approximation Errors

The errors for the second-order approximation increase at a much higher rate than those for the first-order approximation with distance from the expansion point. Figure 2 shows the errors for only the 2020 season, which had the median percentage increase in error among the seasons in the data. Additional plots are provided in Appendix B.

Discussion

The results suggest that the second-order Taylor approximation of the Pythagorean expectation model is inferior to the first-order approximation. The coefficients for the second-order term that resulted from using the method of least squares to fit the second-order model shown in (12) to real MLB data were not significant, and the estimates for the intercept and the first-order coefficients were largely unchanged compared to their counterparts from the first-order model. This not only suggests that the second-order term is unnecessary, but also that the first-order model already adequately explains the data.

The second-order approximation also proved to be inferior to the first-order approximation in terms of its accuracy in approximating the Pythagorean expectation model. The approximation error of the second-order approximation increases at a much

higher rate than that of the first-order approximation with distance from the expansion point. This behavior comes from the second-order term's dependency on the squared distance from the expansion point $(R - R_{\text{avg}})^2 + (RA - R_{\text{avg}})^2$, whereas the first-order term has no such dependency and only depends on the run differential $R - RA$. Therefore, while the second-order approximation was expected to be more accurate to the Pythagorean expectation model, it is less so when applied to the full range of MLB data.

Although the results suggest that the first-order model adequately explains real MLB data, exploring the third-order approximation may still be worthwhile. The S-shaped nature of the Pythagorean expectation curve shown in Figure 1 suggests that a third-order term may correct some of the underestimation by the second-order approximation. An investigation into the validity of this theory and its applicability to real MLB data could prove useful. Another area of inquiry is the variation in γ from season to season. A study could be done on the relationship between γ and factors that change the run environment, such as the baseball itself, league rules, and offensive and defensive approaches. Finally, an examination of alternative second-order terms, such as the square of the run differential $(R - RA)^2$, could be performed to see if another such term could be used to improve the accuracy or explainability of the first-order model.

References

- Dayaratna, Kevin D., and Steven J. Miller. 2012. "First Order Approximations of the Pythagorean Won-Loss Formula for Predicting MLB Teams' Winning Percentages." *By the Numbers: The Newsletter of the SABR Statistical Analysis Committee* 22(1):15–19.
- Friendly, Michael, Chris Dalzell, Martin Monkman, Dennis Murphy, Vanessa Foot, Justeena Zaki-Azat, Daniel J. Eck, and Sean Lahman. 2026. *Lahman: Sean 'Lahman' Baseball Database*. R package version 14.0-0.
<https://CRAN.R-project.org/package=Lahman>.
- MacAree, Graham. 2010. "Pythagorean Win-Loss." *FanGraphs Library*.
<https://library.fangraphs.com/principles/expected-wins-and-losses/>.

Appendix A: Gamma Estimates (using Nonlinear Least Squares)

season	$\hat{\gamma}$	95% Lower Bound on $\hat{\gamma}$	95% Upper Bound on $\hat{\gamma}$
2006	1.953	1.626	2.284
2007	1.678	1.359	1.999
2008	1.932	1.621	2.247
2009	1.956	1.603	2.312
2010	1.638	1.488	1.790
2011	1.847	1.584	2.113
2012	1.976	1.721	2.233
2013	1.757	1.545	1.972
2014	1.810	1.498	2.124
2015	1.711	1.387	2.040
2016	1.691	1.383	2.004
2017	1.613	1.401	1.827
2018	1.743	1.508	1.982
2019	1.941	1.765	2.119
2020	1.861	1.577	2.151
2021	1.761	1.513	2.014
2022	1.715	1.562	1.870
2023	1.690	1.448	1.936
2024	1.793	1.610	1.979
2025	1.629	1.368	1.895

Appendix B: Approximation Errors for the Season with the Smallest Percent Increase (2011) and the Largest Percent Increase (2008) from the First-Order Model to the Second-Order Model

